

The Law of Portfolio Learnability

Simulation section preview

1 Simulation: a balanced three-factor economy

We study direct portfolio learning in a stock-level economy with three economic factors. The managed spectrum is generated by the characteristic distribution and the Matérn kernel. Numerical kernel rank is distinct from the number of economic factors. Table 1 gives the frozen calibration.

1.1 Characteristics, returns, and the population policy

There are $N = 3G$ stocks and $D = 6$ characteristics. Independent group-level states start from $S_{g,d,0} \sim N(0, 1)$ and evolve as

$$S_{g,d,t+1} = \rho_d S_{g,d,t} + \sqrt{1 - \rho_d^2} \eta_{g,d,t+1}, \quad \eta_{g,d,t+1} \sim N(0, 1).$$

With $U_{g,d,t} = \Phi(S_{g,d,t})$, the three roles in group g have

$$Z_{g,r,d,t} = 2((U_{g,d,t} + r/3) \bmod 1) - 1, \quad r = 0, 1, 2.$$

The same phase shift applies to every coordinate. These population ranks have a continuous uniform distribution on the cube; they are not empirical ranks. Writing $u(z) = (z_1 + 1)/2$, define

$$\beta(z) = \sqrt{c_\beta} \begin{pmatrix} \sqrt{2/3} \cos(2\pi u(z)) \\ \sqrt{2/3} \sin(2\pi u(z)) \\ 1/\sqrt{3} \end{pmatrix}.$$

The trigonometric identities for three equally spaced phases imply $\|\beta(z)\|^2 = c_\beta$ and $B'_t B_t / N = \Gamma_\beta = (c_\beta/3)I_3$ at every date. Independent next-period shocks satisfy

$$F_{t+1} \sim N(\mu_F, I_3 - \mu_F \mu_F'), \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2 I_N), \quad R_{t+1} = B_t F_{t+1} + \varepsilon_{t+1}.$$

Thus $\mathbb{E}[F_{t+1} F_{t+1}'] = I_3$ is a second-moment normalization. The conditional mean and second moment of stock returns are $m_t = B_t \mu_F$ and $S_t = B_t B_t' + \sigma_\varepsilon^2 I_N$.

For the paper's weights $w_t(W) = W(Z_t)/N$, minimizing $\mathbb{E}[(1 - w_t(W)' R_{t+1})^2 \mid \mathcal{F}_t]$ gives $S_t w_t^* = m_t$. The identity

$$(BB' + \sigma^2 I)^{-1} B = B(B'B + \sigma^2 I_3)^{-1}$$

and exact cross-sectional balance therefore yield

$$w_t^* = \frac{1}{N} B_t \left(\Gamma_\beta + \frac{\sigma_\varepsilon^2}{N} I_3 \right)^{-1} \mu_F, \quad W^*(z) = \frac{\beta(z)' \mu_F}{c_\beta/3 + \sigma_\varepsilon^2/N}.$$

The population policy is derived from the return model and the loss; it is not an independently imposed simulation target. With $g = c_\beta/3$,

$$q^* = \frac{g}{g + \sigma_\varepsilon^2/N} \|\mu_F\|^2, \quad Q(W^*) = 1 - q^*, \quad (\text{SR}^*)^2 = \frac{q^*}{1 - q^*}.$$

1.2 Assumption audit and spectrum

Stationary initialization and stable Gaussian AR dynamics give geometric beta-mixing; measurable rank and phase maps and independent next-period shocks preserve this property (E1). For $A_t = K_t/N^2$, the exact noncentral Gaussian quadratic-form MGF verifies

$$\log \mathbb{E} \left[\exp \left(R'_{t+1} A_t R_{t+1} / B_X^2 \right) \mid \mathcal{F}_t \right] \leq \frac{1}{2}, \quad B_X^2 = 4(c_\beta + \sigma_\varepsilon^2).$$

Conditional scalar Gaussian moments give $K_4 = 3$ (E2). The explicit sine, cosine, and constant loading functions have smooth extensions on the bounded cube. They belong to the Sobolev space of order $\nu + D/2 = 9/2$, equivalent to the Matérn-3/2 RKHS, as does their fixed linear combination W^* (E3). The conditional pricing restriction follows from $v'_t(m_t - S_t w_t^*) = 0$ for every admissible v_t (E4).

For the uncentered managed second-moment operator,

$$\frac{\sigma_\varepsilon^2}{N} T_K \preceq \Sigma_{\mathcal{H}_K} \preceq \left(c_\beta + \frac{\sigma_\varepsilon^2}{N} \right) T_K.$$

The lower term is the idiosyncratic contribution; the upper bound follows by Jensen's inequality and $\|\beta\|^2 = c_\beta$. Thus the managed spectrum has the Matérn exponent $b = 1 + 2\nu/D = 3/2$ (E5). The finite-sample audit uses one independent population quadrature with 3,072 nodes, including the within-triplet dependence, and tests the Loewner and ordered-eigenvalue bounds. Log-log fits use the fixed windows 30–200, 60–400, and 120–800. These finite-resolution fits do not prove an asymptotic exponent. The analytic optimal Sharpe is positive and finite and is compared with a 100,000-period stock-return simulation (E6).

1.3 Estimation, selection, and evaluation

We use the paper's response-one ridge estimator in a controlled Nyström subspace of the actual Matérn RKHS, without rescaling the kernel diagonal. The inducing points are independent of all training and OOS samples. Empirical effective complexity is

$$\hat{\mathcal{C}}(\lambda) = \text{Tr}[\hat{\Sigma}_T(\hat{\Sigma}_T + \lambda I)^{-1}].$$

Training sample sizes and all 32 positive penalties are fixed before simulation. Every replication generates genuine stock-level returns. Sample-size paths are nested within a replication; different replications are independent. Validation uses the last quarter of the available history, following an earlier training block. After selecting lambda by validation loss, the policy is refitted on all available observations. An independent 120-period block gives OOS loss and unannualized Sharpe. Neither population moments, W^* , nor OOS data enter selection. Separate independent population quadrature integrates conditional factor and idiosyncratic moments for more precise population-risk evaluation.

The population-risk oracle minimizes mean regret over the same fixed grid. Boundary choices and all high-complexity observations are retained. We report means, medians, pointwise Monte Carlo intervals, selection frequencies, and benchmarks for strong shrinkage, near-unregularized policies, a validated affine characteristic policy, and equal weights.

For each penalty, the exact decomposition in the numerical subspace is

$$Q(\widehat{W}_\lambda) - Q(W^*) = \{Q(W_\lambda) - Q(W^*)\} + \|\widehat{W}_\lambda - W_\lambda\|_\Sigma^2 + 2\langle \widehat{W}_\lambda - W_\lambda, \Sigma W_\lambda - \mu \rangle.$$

The cross term is signed; only the squared-norm estimation term is necessarily nonnegative. Population bias includes the explicitly measured numerical approximation floor. Rank sensitivity compares 128, 256, and 512 inducing points.

1.4 Monte Carlo evidence and limits

Smoke-test preview only; these are not production paper results. The headline uses 4 independent replications. At $T = 240$, mean oracle and validation-selected population losses are 0.999072 and 1.000795; their mean population Sharpes are 0.05547 and -0.00455. Mean empirical complexities are 5.092 and 0.274. The selected-to-oracle mean regret ratio is 1.130. The validation boundary-selection frequency is 25.0 percent. The ensemble oracle selects a grid boundary at 2 of 3 sample sizes. These observations are retained in all plots and rate regressions.

This smoke preview fits only predeclared windows containing at least three sample sizes; the full paper profile additionally uses the upper-half and largest-four windows. Bootstrap intervals resample entire independent replications, preserving their joint T-by-lambda surfaces and recomputing both the oracle and validation choices. The displayed theoretical slopes are benchmarks for the theory’s risk envelope; they are not imposed DGP-specific equalities for this fixed smooth population policy. Slower convergence, boundary selections, and differences from benchmark slopes are reported without modifying the calibration or fitting windows.

The full robustness design is predeclared but is not represented by this smoke-test preview.

Table 1: Baseline calibration and frozen Monte Carlo design.

Parameter	Value
N	600
T	60, 120, 240
D	6
Economic factors	3
ρ	.95
c_β	0.0016
μ_F	(0.1, 0.06, 0.03)
σ_ε	0.08
Matérn ν	1.5
Length scale	1
Theoretical b	1.5
Population Sharpe	0.12008616
Nystrom rank	64
Headline replications	4
Independent OOS length	120

Table 2: Baseline assumption audit. The spectral band is a finite-resolution diagnostic.

Condition	Theoretical status	Diagnostic	Tolerance	Result
E1	Exact: stationary stable Gaussian AR	spectral radius 0.950	< 1	Pass
E2(a)	Exact Gaussian quadratic MGF	max log MGF 0.0040657	≤ 0.5	Pass
E2(b)	Conditional scalar Gaussian	max ratio 3	≤ 3	Pass
E3	Smooth beta in Sobolev $H^{\hat{}}(9/2)$	fixed finite linear combination	analytic	Pass
E4	Exact conditional normal equation	max residual 1.47e-17	$< 1e-11$	Pass
E5	Loewner sandwich; theoretical b=1.5	slope -1.2399	-1.5 +/- 0.35	Pass
E6	Analytic optimal Sharpe	MC 0.12297; exact 0.12009	abs error ≤ 0.0190	Pass

Table 3: Monte Carlo mean population performance. Validation uses no population information; the oracle minimizes mean population regret over the fixed grid.

T	Oracle loss	Selected loss	Oracle SR	Selected SR	Oracle C	Selected C	Oracle lambda	Selected lambda	Regret ratio
60	0.999998	1.130372	0.00011	0.00422	0.0112	9.096	0.01	0.0002566	10.17
120	1.000019	1.000762	0.01227	0.01181	0.01117	0.09554	0.01	0.007561	1.052
240	0.999072	1.000795	0.05547	-0.00455	5.092	0.2744	1.346e-06	0.003831	1.13

Table 4: Fixed-window rate regressions with whole-replication bootstrap intervals. Theoretical slopes are benchmarks, not fitted targets.

quantity	window	theoretical_slope	OLS_slope	OLS_SE	bootstrap_low	bootstrap_high
oracle_lambda	full	-0.6	-6.43	3.712	-6.43	1.072
oracle_lambda	largest_four	-0.6	-6.43	3.712	-6.43	1.072
oracle_complexity	full	0.4	4.398	2.539	-1.052	4.398
oracle_complexity	largest_four	0.4	4.398	2.539	-1.052	4.398
oracle_regret	full	-0.6	-0.04855	0.02929	-0.4096	0.001818
oracle_regret	largest_four	-0.6	-0.04855	0.02929	-0.4096	0.001818
oracle_sharpe_gap	full	-0.3	-0.4464	0.1688	-0.7006	0.04052
oracle_sharpe_gap	largest_four	-0.3	-0.4464	0.1688	-0.7006	0.04052
selected_lambda	full	-0.6	1.95	1.692	1.616	4.411
selected_lambda	largest_four	-0.6	1.95	1.692	1.616	4.411
selected_complexity	full	0.4	-2.636	2.423	-2.694	-2.121
selected_complexity	largest_four	0.4	-2.636	2.423	-2.694	-2.121
oracle_empirical_complexity	full	0.4	4.414	2.551	-1.095	4.479
oracle_empirical_complexity	largest_four	0.4	4.414	2.551	-1.095	4.479
selected_empirical_complexity	full	0.4	-2.526	2.337	-2.577	-2.123
selected_empirical_complexity	largest_four	0.4	-2.526	2.337	-2.577	-2.123
selected_regret	full	-0.6	-1.634	0.9452	-2.237	-0.2497
selected_regret	largest_four	-0.6	-1.634	0.9452	-2.237	-0.2497
selected_sharpe_gap	full	-0.3	0.05265	0.08687	-0.0408	0.2091
selected_sharpe_gap	largest_four	-0.3	0.05265	0.08687	-0.0408	0.2091

Table 5: Robustness at the largest sample size. Heteroskedastic regret uses a finite-subspace reference; it is not a theorem-baseline result.

Environment	N	rho	nu	Oracle regret	Selected regret	Oracle C	Selected C	Selected SR	Boundary frequency
baseline	600	0.95	1.5	0.01329	0.01501	5.092	0.2744	-0.004555	0.25

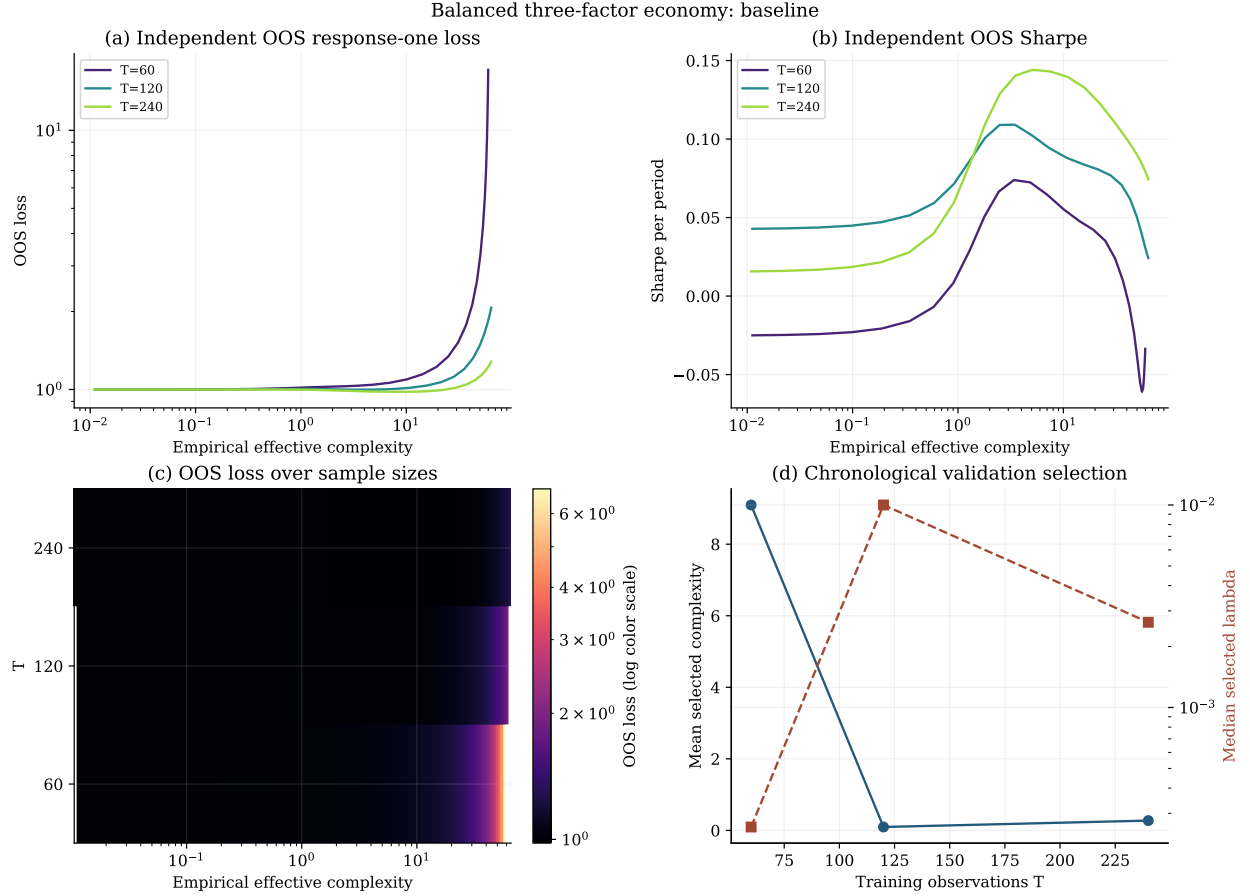


Figure 1: Independent OOS performance against empirical effective complexity. All penalties are retained; loss uses a logarithmic scale.

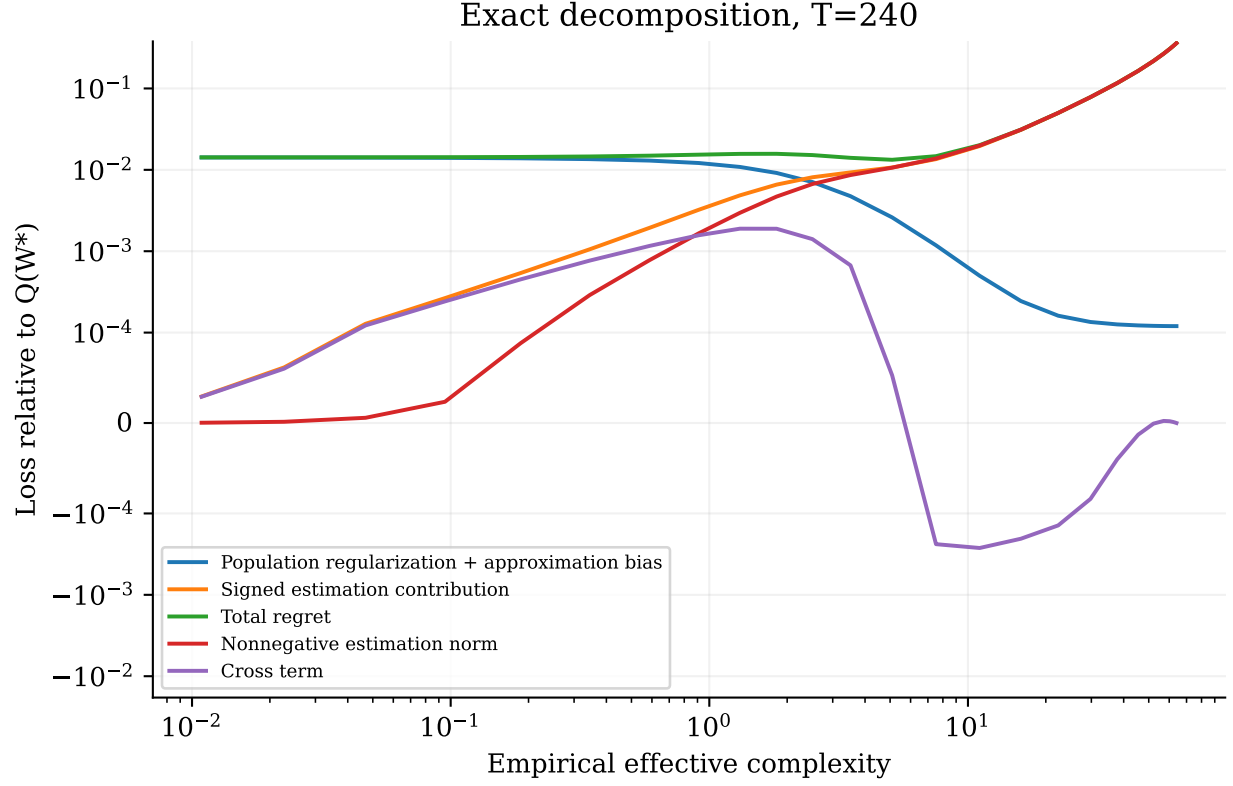


Figure 2: Exact bias, signed estimation contribution, and total regret. A symmetric logarithmic scale retains negative cross terms.

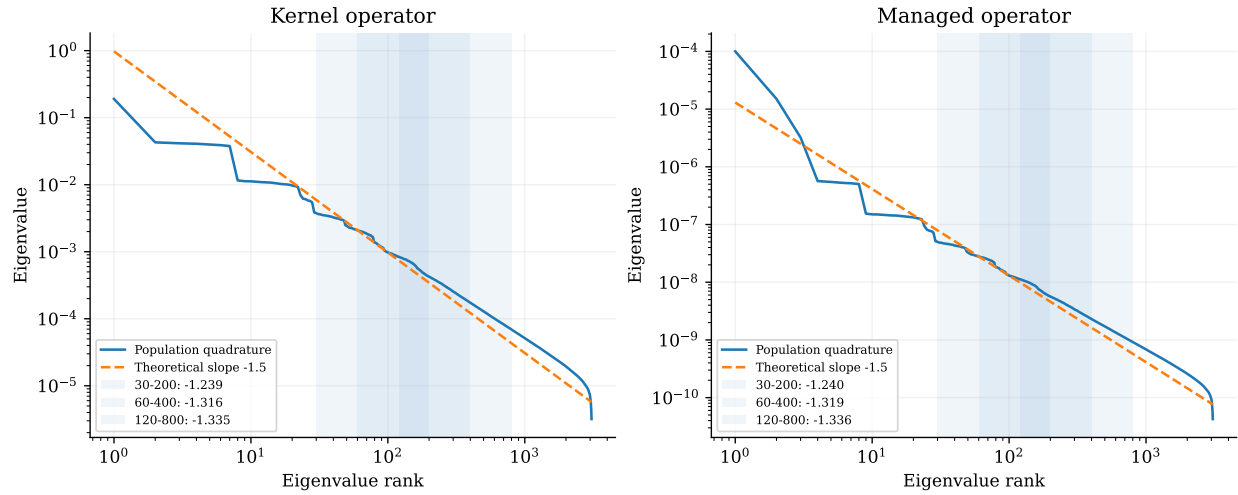


Figure 3: Population kernel and managed spectra on common independent quadrature. Shaded fixed windows show spectral-fit sensitivity.

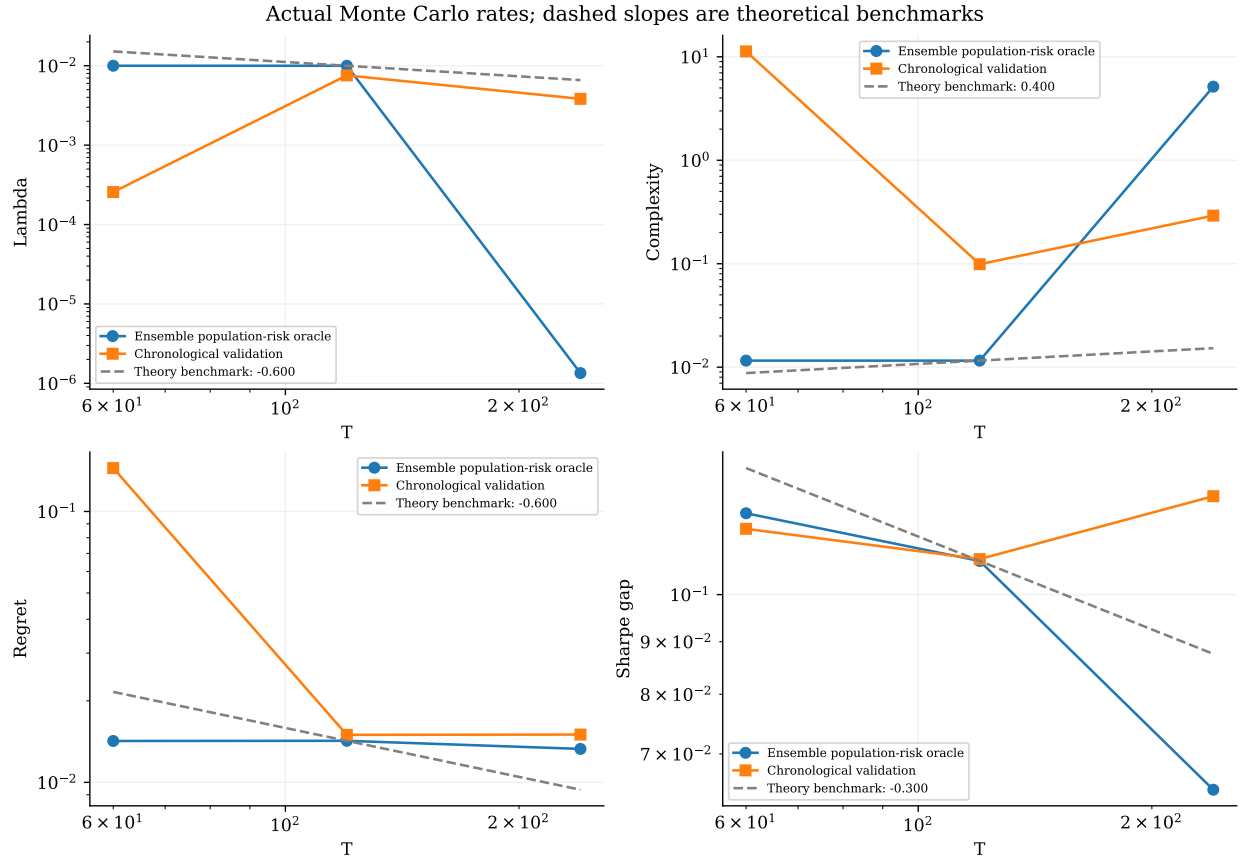


Figure 4: Oracle and validation rate paths. Dashed curves display theoretical slope benchmarks without constraining fitted results.

Simulation appendix

Population evaluation uses independent group quadrature. In orthonormal Nyström coordinates $\phi(z)$, write $H_g = \frac{1}{3} \sum_r \phi(Z_{g,r}) \beta(Z_{g,r})'$ and $\bar{H} = M^{-1} \sum_g H_g$. Then

$$\hat{\Sigma} = \frac{\sigma_\varepsilon^2}{N} \hat{T}_K + \frac{1}{GM} \sum_g H_g H_g' + \left(1 - \frac{1}{G}\right) \bar{H} \bar{H}', \quad \hat{T}_K = \frac{1}{3M} \sum_{g,r} \phi(Z_{g,r}) \phi(Z_{g,r})'.$$

This integrates the factor and idiosyncratic shocks analytically while preserving within-triplet dependence. The quadrature distribution is independent of all training paths and cannot affect validation choices. For empirical ranks, full independent cross-sections are integrated because ranking couples groups. For heteroskedasticity, the idiosyncratic term uses the actual bounded state-dependent conditional variance.

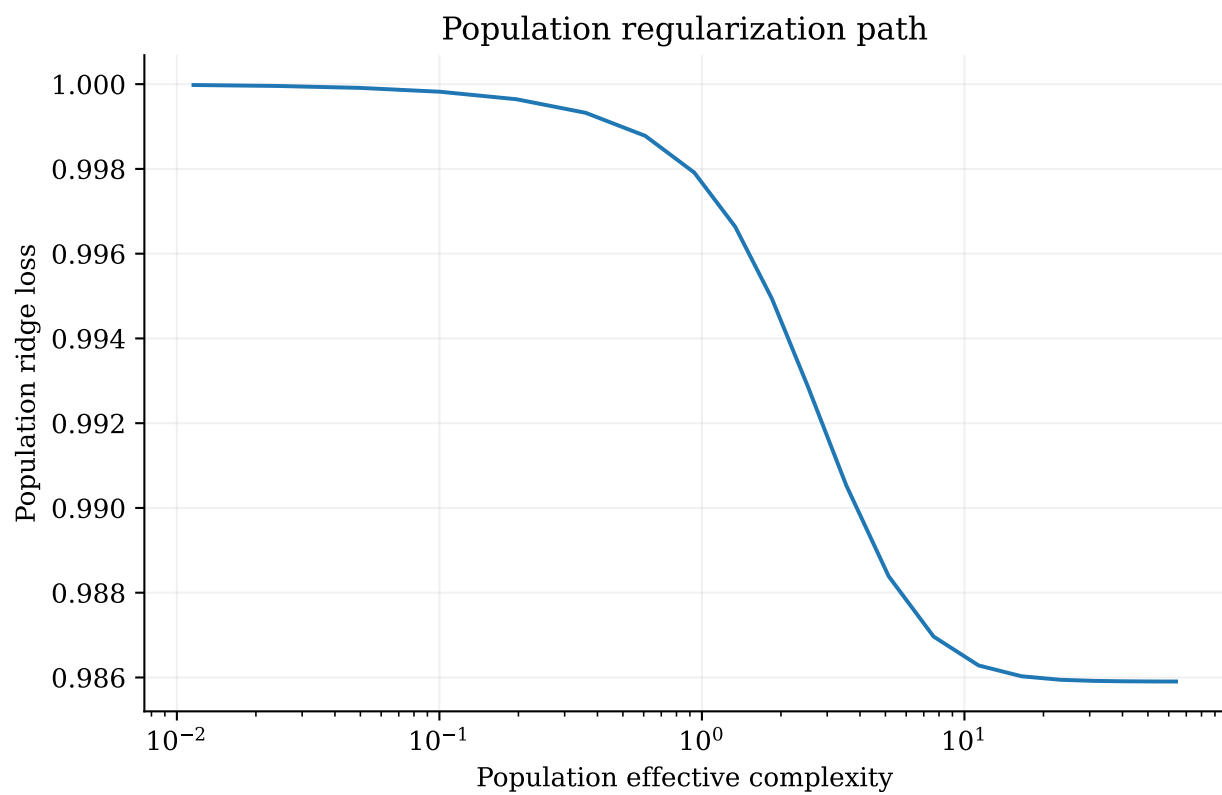


Figure 5: Population ridge loss versus population effective complexity.

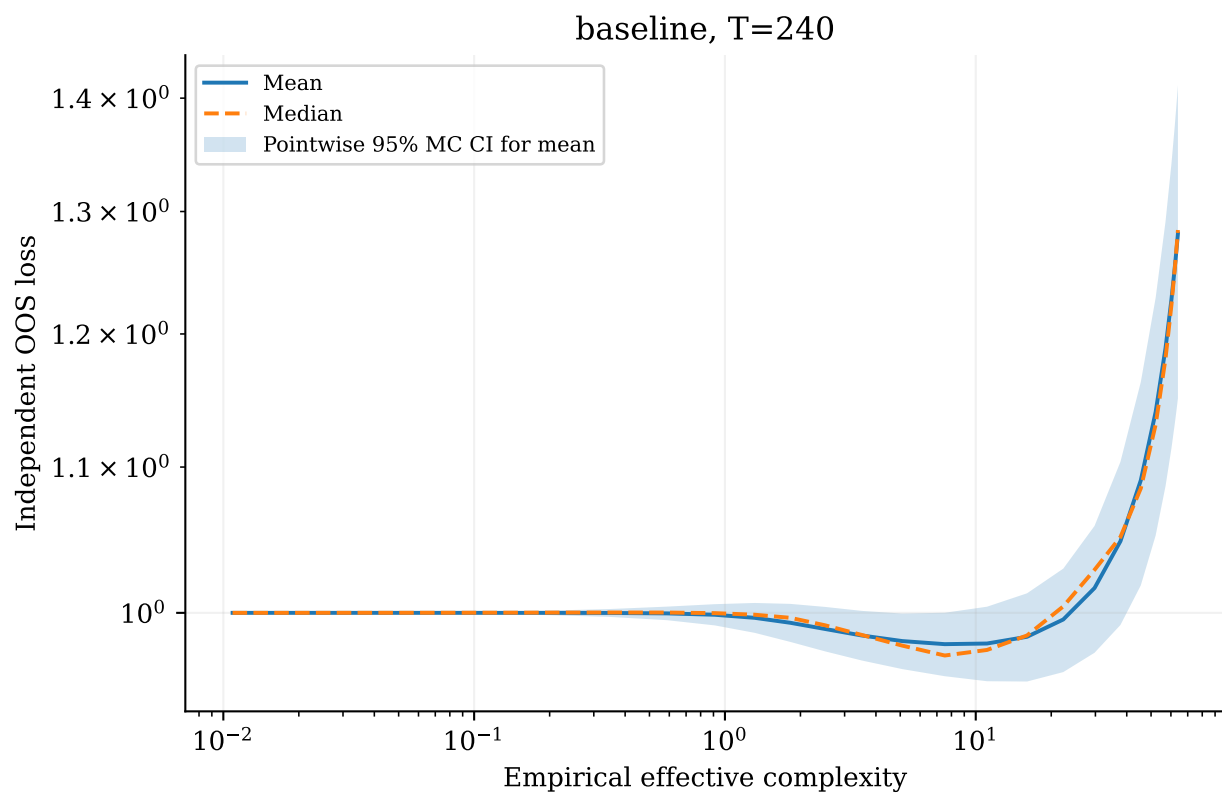


Figure 6: OOS loss: Monte Carlo mean, median, and pointwise confidence interval for the mean.

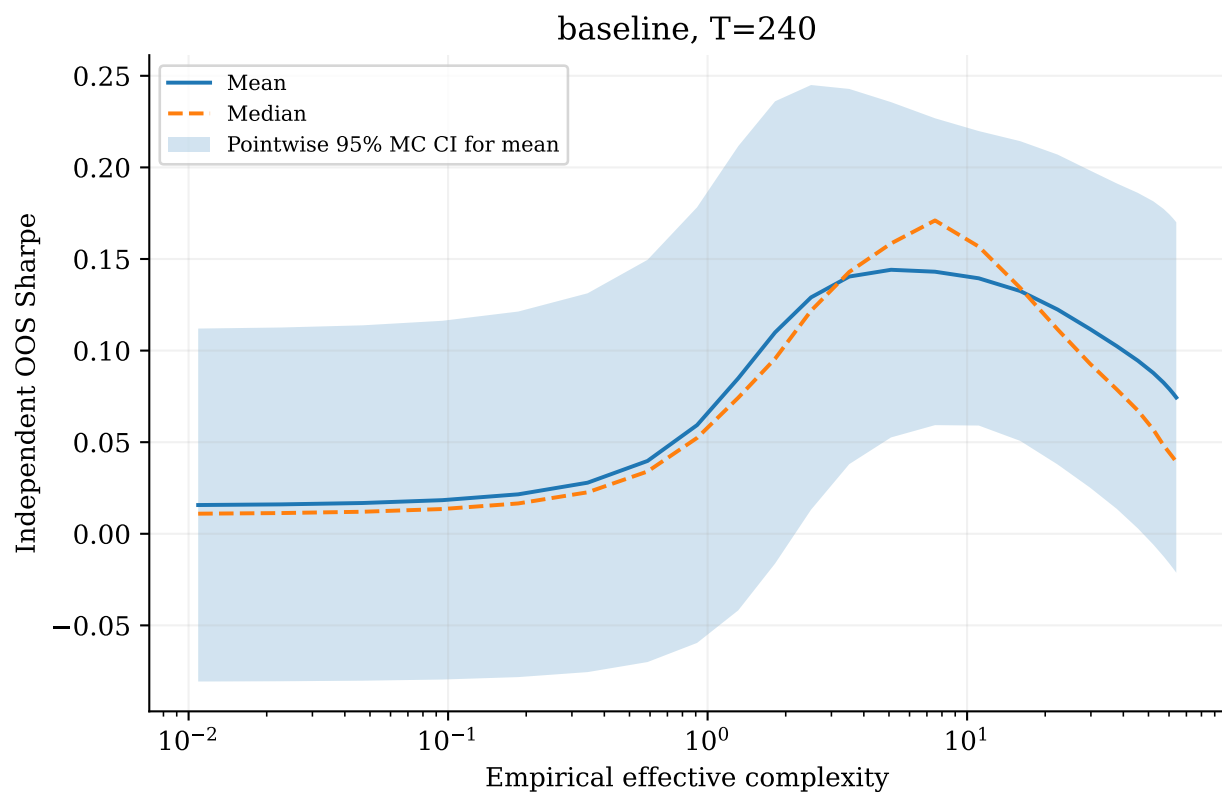


Figure 7: OOS Sharpe: Monte Carlo mean, median, and pointwise confidence interval for the mean.

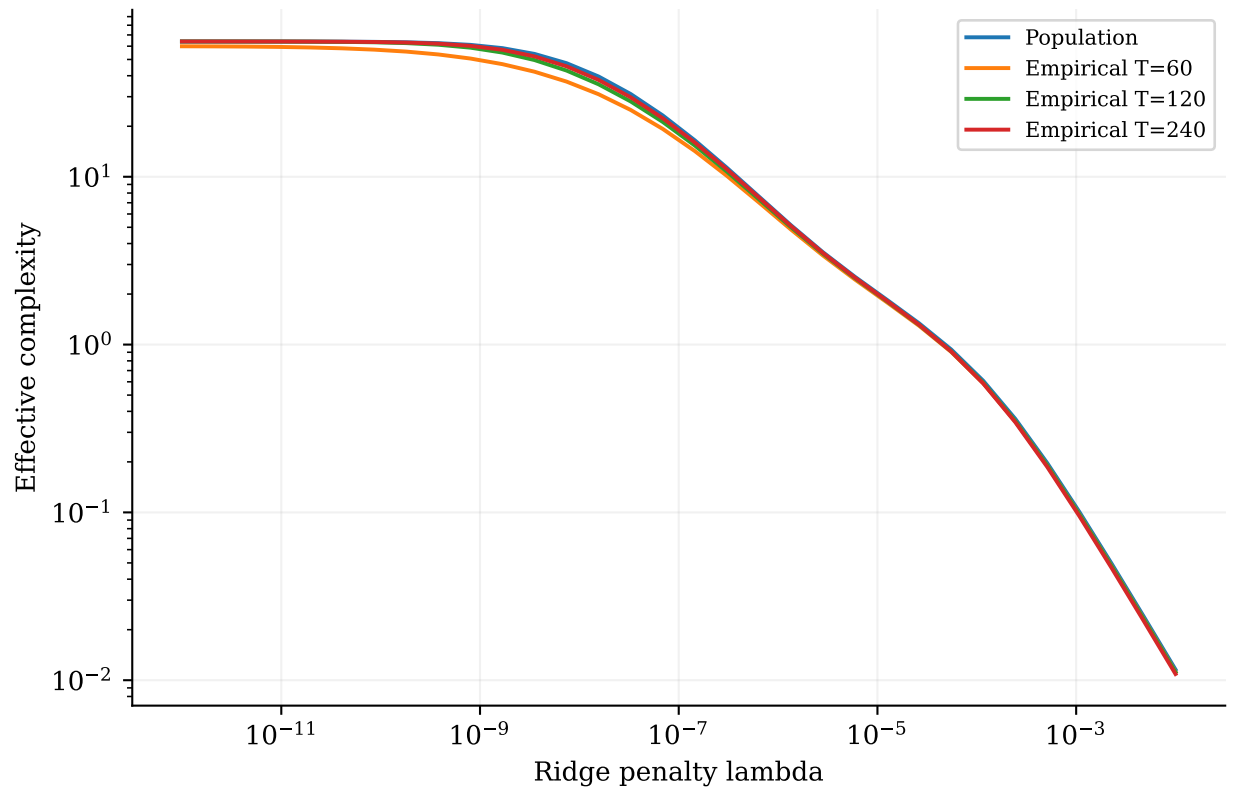


Figure 8: Population and empirical effective complexity as reparameterizations of regularization.

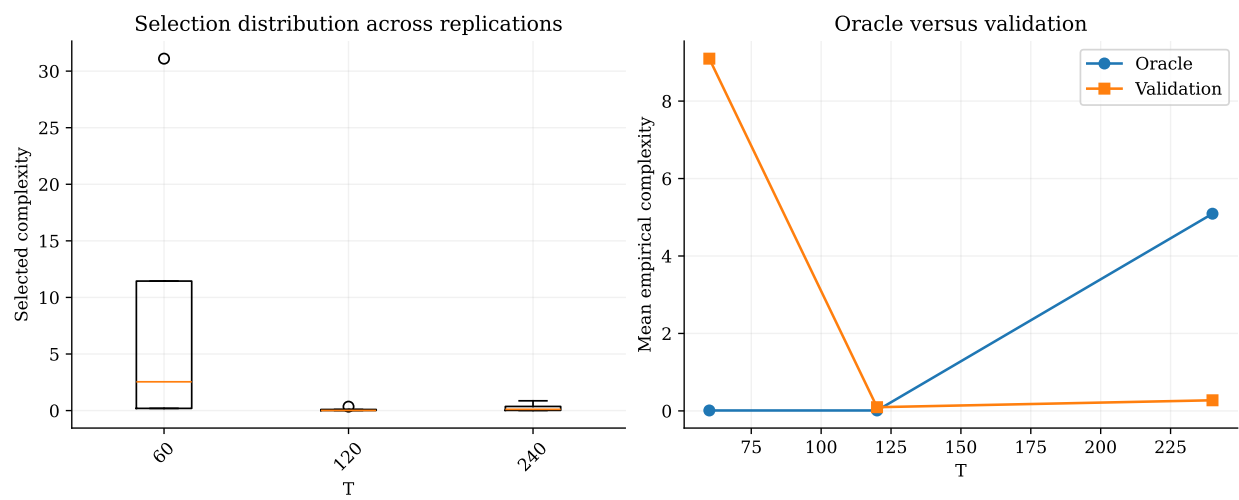


Figure 9: Selection distributions and oracle versus selected empirical complexity.